CMSC424: Database Design
Module: Query Languages

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Learning Goals

- Key abstraction: A set or multi-set of objects or key-value pairs or tuples, called a “dataset”
- Basic operations on datasets
  - Take one or more datasets as input, produce a dataset as output
- Different ways to specify the above operations in different systems
  - Relational Algebra, SQL, MongoDB, Apache Spark (basics)
- Advanced SQL features
  - Focus on what’s possible, less on syntax
- Homework 1 (& Quiz 1) on the first three topics, Programming Assignment 1 covers last topic

Inputs and Outputs: Datasets

- Sets or "multi-sets" or "collections" of ..
  - Objects → key-value pairs → tuples

o1
o2
o3
o4

Set of objects (e.g., images, JSON documents)

k1	o1
k2	o2
k3	o3
k4	o4

Set of key-value pairs (e.g., pairs of (userid, profile))

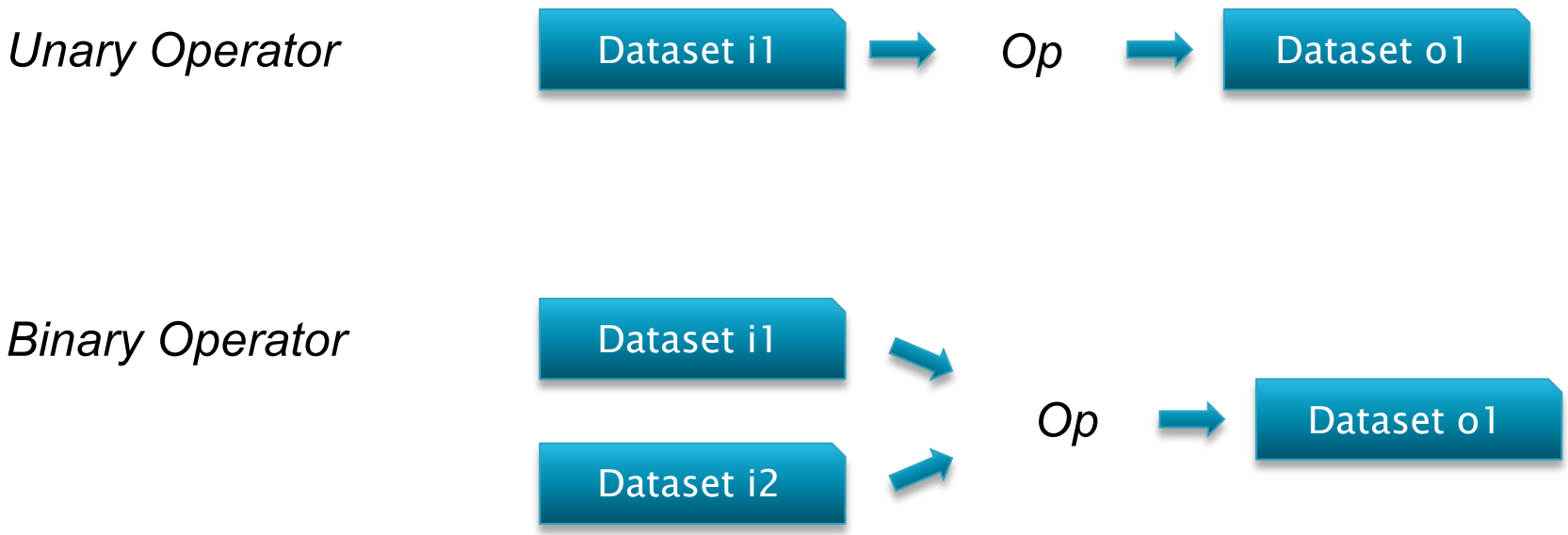
o1
o1
o1
o4

Multi-set of objects

A	B	C	D
α	α	1	7
α	β	5	7
β	β	12	3
β	β	23	10

Set of tuples aka Relation/Table

Operations on Datasets



- An operation involving “objects” typically requires some explicit code (e.g., in python) to interprete it
  - i.e., the ”schema”/”structure” is implicit in the code
- An operation that works on a collection of “objects” also works on a collection of ”key-value pairs” or “tuples”
- Opeartions with more than 2 inputs possible, but less common

Select/Find/Match/Filter

- Input: Object Collection, Output: Object Collection
  - So also works on Relations
- Select only those objects that match the condition
  - Need a predicate or a "function" that returns a boolean

Relation r

A	B	C	D
α	α	1	7
α	β	5	7
β	β	12	3
β	β	23	10

Select tuples where
A = B and D > 5

A	B	C	D
α	α	1	7
β	β	23	10

Select/Find/Match/Filter

- Input: Object Collection, Output: Object Collection
  - So also works on Relations
- Select only those objects that match the condition
  - Need a predicate or a “function” that returns a Boolean

Input

o1	
o2	
o3	
o4	

Select images with cats

o1	
o2	
o4	

Need a function of the form:

```
boolean has_a_cat(image_data)
```

Select/Find/Match/Filter: Language Constructs?

- Relational Algebra (only on relations)
  - Symbol: σ (sigma)
- SQL
  - Uses "where" clause

Relation r

A	B	C	D
α	α	1	7
α	β	5	7
β	β	12	3
β	β	23	10

$\sigma_{A=B \wedge D > 5}(r)$

A	B	C	D
α	α	1	7
β	β	23	10

```
select *
from r
where r.A = r.B and r.D > 5
```

"r." optional if no ambiguity

Select/Find/Match/Filter: Language Constructs?

- Apache Spark (and many other similar frameworks)
  - Use “filter” operation
  - No explicit schema though, so need to provide code

Input I

o1	
o2	
o3	
o4	

I.filter(lamba x: has_cat(x))

Output O

o1	
o2	
o4	

Interpretation:

```
for each object x in I:
    if has_cat(x):
        O.append(x)
```

Select/Find/Match/Filter: Language Constructs?

- Apache Spark (and many other similar frameworks)
  - Use “filter” operation
  - No explicit schema though, so need to provide code

Input I: key-value pairs

k1	o1
k2	o2
k3	o3
k4	o4

I.filter(lamba x: has_cat(x[1]))

Output O

k1	o1
k2	o2
k4	o4

Interpretation:

```
for each object x in I:
    if has_cat(x[1]):
        O.append(x)
```